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Problem Solving Protocol

MRAgent: an LLM-based automated agent for causal knowledge discovery in disease via Mendelian randomization

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Abstract

Understanding causality in medical research is essential for developing effective interventions and diagnostic tools. Mendelian Randomization (MR) is a pivotal method for inferring causality through genetic data. However, MR analysis often requires pre-identification of exposure-outcome pairs from clinical experience or literature, which can be challenging to obtain. This poses difficulties for clinicians investigating causal factors of specific diseases. To address this, we introduce MRAgent, an innovative automated agent leveraging Large Language Models (LLMs) to enhance causal knowledge discovery in disease research. MRAgent autonomously scans scientific literature, discovers potential exposure-outcome pairs, and performs MR causal inference using extensive Genome-Wide Association Study data. We conducted both automated and human evaluations to compare different LLMs in operating MRAgent and provided a proof-of-concept case to demonstrate the complete workflow. MRAgent's capability to conduct large-scale causal analyses represents a significant advancement, equipping researchers and clinicians with a robust tool for exploring and validating causal relationships in complex diseases. Our code is public at https://github.com/xuwei1997/MRAgent.

Keywords: MRAgent; AI agent; causal knowledge discovery; Mendelian randomization; large language models (LLMs)

Introduction

Understanding causality is of paramount importance in medical research, as it forms the foundation for developing effective prevention strategies, diagnostic tools, and therapeutic interventions [1]. Establishing causal relationships between exposures and health outcomes enables researchers to identify risk factors, understand disease mechanisms, and predict the potential impact of interventions. Without a clear grasp of causal links, medical research risks being confounded by mere associations, which may lead to ineffective or even harmful clinical practices.

Mendelian Randomization (MR) [2] has emerged as a powerful and mainstream tool for causal inference in medical research, leveraging genetic variation to assess whether modifiable exposures affect health outcomes. This technique is grounded in the principle that genetic variants, which are randomly assorted during gamete formation, can serve as instrumental variables to infer causality, akin to the randomization in controlled trials [3]. Typically, MR analysis requires the pre-identification of exposure-outcome pairs, often derived from clinical experience or existing literature. However, in many scenarios, these pairs are not readily available, posing a challenge for clinicians who need to swiftly investigate the causal factors or consequences of specific diseases. Although some studies have attempted large-scale MR analyses, the vast number of potential diseases and influencing factors makes exhaustive MR research through brute force approaches impractical.

On the other hand, while numerous studies have examined causality in the context of diseases, many are limited to providing statistical validation for specific exposure-outcome pairs or merely documenting isolated case studies. Although these studies contribute valuable data, they often fall short of delivering definitive causal conclusions, leading to ambiguity in understanding disease mechanisms and impeding the development of targeted interventions. This highlights the need for more rigorous causal inference methodologies. The exposure-outcome pairs identified in traditional observational studies can serve as valuable prior knowledge for MR analyses. Through causal inference analysis based on genetic data, we strengthen our understanding of the exposure-outcome relationships.

Building on the aforementioned insights, a comprehensive pipeline can be established to facilitate knowledge discovery from existing literature, extracting exposure and outcome pairs for subsequent MR causal inference. Such a pipeline would

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significantly enhance medical research by providing clinicians with a systematic approach to uncovering causal relationships, particularly in addressing complex and rare diseases. However, several challenges persist. The sheer volume of literature related to any given disease can be overwhelming, making it labor-intensive to read, comprehend, and accurately extract relevant exposure-outcome pairs. Additionally, it is necessary to determine whether these pairs have already been analyzed through MR or validated via randomized controlled trials. Furthermore, although standardized MR analysis pipelines, such as the TwoSampleMR R package [3, 4], are available, they still require manual R script writing and various labor-intensive tasks. These tasks include searching for suitable Genome-Wide Association Study (GWAS) data [5] and interpreting complex results, which demand significant manual effort and computer programming skills that many clinicians lack. Consequently, there is a pressing need for an automated, user-friendly tool to complete the knowledge discovery-causal inference process.

The rapid advancement of Large Language Models (LLMs) [6, 7] has led to human-level intelligence across numerous tasks. Recently, the emergence of LLM-based agents has garnered substantial attention within the research community, encompassing essential components such as brain, perception, and action [8]. Analogous to human capabilities, these AI agents possess the ability to acquire information from the external environment, make informed decisions, and execute actions utilizing tools [9–13]. The conceptualization of AI agents has paved the way for establishing a more intelligent pipeline for causal knowledge discovery in diseases and automated MR analysis.

In this paper, we introduce MRAgent, an LLM-based Automated Agent for Causal Knowledge Discovery in Disease via MR. Upon inputting a specific disease, MRAgent autonomously scans and analyzes relevant literature from PubMed [14] to identify potential exposures or outcomes associated with the disease. It then records exposure-outcome pairs that have already undergone causal analysis. For each pair that has not been analyzed causally, MRAgent performs MR analysis on multiple sets of GWAS data, ultimately generating a comprehensive analysis report for each exposure-outcome pair. This innovation effectively automates the process of causal knowledge discovery, offering significant advantages to medical research.

Specifically, the contributions of this paper are outlined as follows:

- We have designed the overall architecture of MRAgent, where its brain is powered by Large Language Models (LLMs) for information extraction and process control, with data tables serving as its memory. A toolkit is provided for perception and action, such as acquiring PubMed article data and performing two-sample MR analysis.
- We have deconstructed the traditional MR study into several essential steps, offering a structured approach that not only clarifies the process but also serves as a valuable reference for automating MR analysis.
- We have constructed MRAgent's workflow based on traditional MR processes to enable automated knowledge discovery and causal inference.
- To ensure the reliability of MRAgent, we designed comprehensive tests for each step involving LLMs, which included subjective evaluations by experts and similarity assessments between generated outputs and reference standards using SimCSE [15]. Additionally, we experimented with different prompt strategies and analyzed the execution time for various processes.
- We provide a complete proof-of-concept case to demonstrate the efficacy of MRAgent in real-world applications.

Materials and methods

The MR Agent is a sophisticated system structured according to the paradigm proposed by Xi [8], as illustrated in Fig. 1. The Agent's brain comprises Knowledge, Analysis, and Process Control, implemented using LLMs, while Memory is maintained through Data Sheet files. For perception and action, the Toolkit facilitates the acquisition of external information and the execution of MR computations. By emulating the traditional MR analysis workflow and integrating these components, we have constructed an intelligent MRAgent workflow. Researchers can input exposures and/or outcomes for study, and MRAgent will automatically conduct the MR analysis, producing a comprehensive research report. This section details the components that implement the brain, perception, and action, and introduces the complete workflow of MRAgent.

Brain

LLM-based knowledge, analysis & process control

MR research involves a decision-making process that is both flexible and complex, requiring the Agent's brain to possess extensive biomedical knowledge and the ability to perform sophisticated analysis and process control by integrating results from various steps with domain-specific insights. State-of-the-art (SOTA) LLMs exhibit human-level intelligence, making them exceptionally suitable for these challenges [8]. We designed MRAgent to act as a biomedical expert in MR, using system prompts like "You are a helpful biomedical scientist."

MRAgent processes initial inputs and tool invocation results through carefully designed prompts. The LLM generates structured outputs, stored in data sheets (as detailed in *Data sheet memory section*), and Python scripts parse these data to execute operations based on their content, thus enabling the LLM to control the overall workflow. The primary objective of MRAgent is to generate comprehensive reports, with the LLM performing an extensive analysis of MR results, ensuring the final output is both insightful and reliable.

Data sheet memory

Memory is a crucial component of the Agent's brain, enabling it to retain and utilize information effectively throughout the MR analysis process. We have designed three specific data sheets that serve as MRAgent's memory, stored in CSV format for ease of access and manipulation.

The first data sheet, `exposure_and_outcome`, records paired exposure and outcome information, including research titles, previous MR analyses, and available GWAS data. The second data sheet, `outcome`, captures information on individual outcomes or exposures, including a comprehensive list of GWAS IDs. The third data sheet, `run`, logs the final pairs of exposures and outcomes selected for MR analysis. The generation and population of these tables will be discussed in detail in *MRAgent workflow section*.

These data sheets adhere to the normal forms of database design [16], ensuring efficient data organization and query capability without redundancy. MRAgent systematically records the results of each step into the Data Sheet Memory, retrieves necessary parameters for subsequent steps, and allows human researchers to review and refine contents, ensuring precise workflow control. This meticulous logging and data-driven decision-making ensure a robust and adaptable MR analysis process.

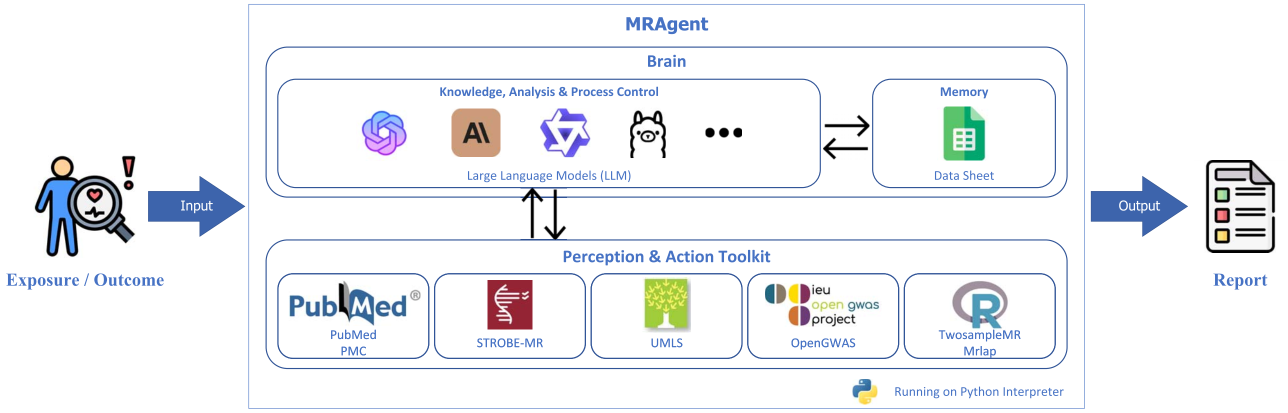

Figure 1. MRAgent architecture.

Perception and action toolkit

MRAgent's LLM-based Brain employs a comprehensive toolkit to gather external information and perform MR computations. This toolkit includes access to the PubMed Medical Citation Database and PubMed Central (PMC) for retrieving scientific literature [14]. While PubMed provides access to article titles, abstracts, and keywords, PMC offers full-text articles, which are essential for in-depth analyses, such as evaluating the quality of published MR studies using the STROBE-MR checklist [17].

The OpenGWAS Summary Datasets provide genome-wide association study data and operate in both "csv" and "online" modes for flexible access [5]. The toolkit also incorporates UMLS Terminology Services for precise medical terminology matching [18].

For MR analyses, the TwoSampleMR Tool [3, 4] is employed, offering both classical MR methods [2] and the MR-MOE method, which utilizes a mixture-of-experts system [19]. Additionally, the MRlap package [20] extends the capabilities of TwoSampleMR by allowing analyses with potentially overlapping samples, correcting for biases such as sample overlap and weak instruments.

These resources collectively enable MRAgent to efficiently conduct causal inference by integrating comprehensive data and advanced analytical methods. Details of each tool in the Perception and Action Toolkit can be found in Appendix I, which provides an in-depth explanation of their functionalities and applications within the MRAgent framework.

Workflow

Traditional MR analysis workflow

To construct an intelligent MR Agent, it is crucial to first understand the traditional workflow of conducting a two-sample MR analysis using the R package TwoSampleMR, As shown in Fig. 2(a).

Step 1: Identifying Exposure and Outcome

The initial step in MR analysis is to identify the relevant exposure and outcome. As discussed in the Introduction, these can be derived from existing literature that reports simple correlations or independent cases. They can also be drawn from clinical observations and practical experience. This step lays the foundation for subsequent analysis to establish potential causal relationships.

Step 2: Checking Previous MR Analyses

Researchers then review published articles to determine if the selected exposure and outcome have been previously subjected to causal inference. If prior studies exist, their conclusions can directly inform clinical practice, avoiding redundant analysis.

Step 3: Identifying Synonyms

Identify all medically relevant synonyms for the exposure and outcome to ensure accurate mapping of terms to corresponding GWAS IDs in subsequent steps. Successfully identifying these synonyms typically requires researchers to have a deep understanding of the disease in question, being well-versed in the various terminologies and synonyms used in medical literature and clinical practice. In practical analysis, this step is often integrated with Step 4, where different terms are substituted based on experience while searching the GWAS database. We list it separately here for clarity in building MRAgent.

Step 4: Selecting the GWAS ID

Using exposure, outcome, and their synonyms as keywords, researchers search the OpenGWAS database. The search results are based on direct keyword matches, which may not always be entirely accurate. Researchers must rely on their understanding of the disease to determine the appropriate GWAS IDs by examining relevant columns such as trait descriptions. For example, a search for "BMI" in OpenGWAS may return datasets for "Body mass index (BMI)" and "Waist-to-hip ratio adjusted for BMI." In this case, the dataset corresponding to "Body mass index (BMI)" should be selected. Additionally, when selecting GWAS IDs, it is crucial that the exposure and outcome maintain the same population.

Step 5: MR Analysis and Interpreting the Results

Once relevant datasets are identified, they can be given to the TwoSampleMR package to conduct the MR analysis, using statistical tools to assess the causal relationship between exposure and outcome. Researchers analyze the computed results to draw conclusions about the causal relationship between the exposure and outcome.

It is important to note that the research process may require repeating several of the aforementioned steps multiple times. This iterative approach ensures the robustness and validity of the research findings, ultimately leading to the conclusion of the study.

MRAgent workflow

MRAgent is a ReACT-based [21] agent designed for knowledge discovery by extracting exposure-outcome pairs from existing correlation and case studies, followed by systematic and large-scale causal inference using MR. Inspired by traditional MR methods, MRAgent leverages LLMs and external toolkits to establish a novel workflow for causal knowledge discovery, as illustrated in Fig. 2(b). To improve the reproducibility and transparency of the workflow, we have documented the prompts used in the process in Appendix II. The detailed workflow construction is as follows:

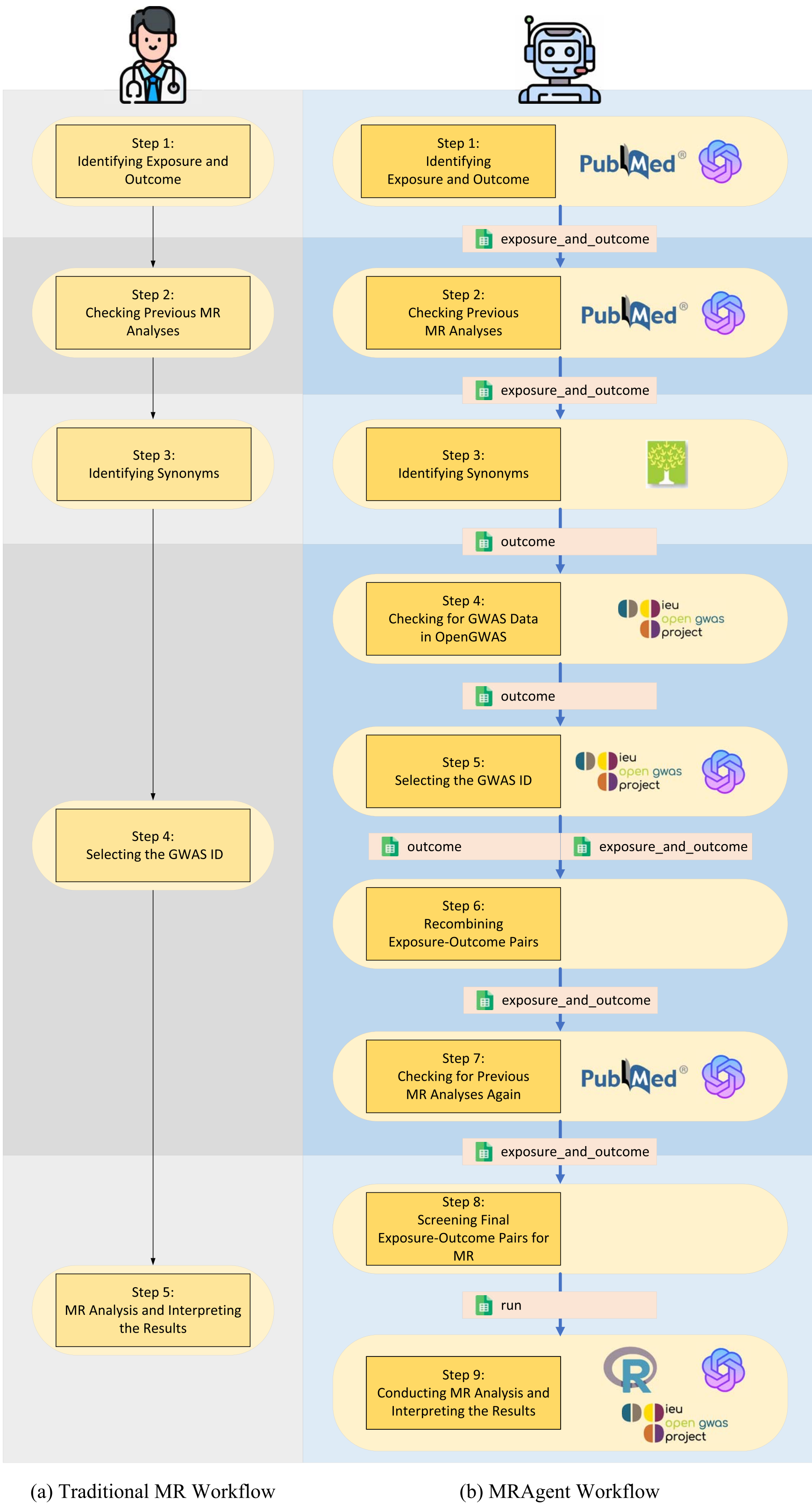

Figure 2. Workflow.

Step 1: Identifying Exposure and Outcome

Initially, MRAgent utilizes the PubMed Medical Citation Database tool to extract exposure-outcome pairs in bulk from relevant literature. The LLM processes these papers to identify and save potential exposure and outcome pairs in a Data Sheet Memory in a predefined format. The number of papers reviewed can be specified during the initialization of MRAgent, and upon completion of this step, the `exposure_and_outcome` table will contain multiple pairs for further analysis.

Additionally, MRAgent allows researchers to input one or more exposure and outcome pairs based on their prior knowledge. These pairs are directly saved into the `exposure_and_outcome` table, effectively making MRAgent a fully automated MR executor for these predefined pairs.

In MRAgent's implementation, discovering exposure-outcome pairs from literature is defined as the "Knowledge Discovery" mode, while directly inputting exposure-outcome pairs is defined as the "Causal Validation" mode. The distinction between these modes lies solely in this initial step.

Step 2: Checking Previous MR Analyses

In line with traditional MR analysis, the agent examines the `exposure_and_outcome` table and searches for relevant papers using the PubMed Medical Citation Database tool with the keywords "(Exposure [Title/Abstract]) AND (Outcome [Title/Abstract]) AND (MR [Title/Abstract])." If an MR analysis has already been conducted for a given exposure-outcome pair, the relevant papers are identified. The LLM reads these papers to provide a definitive "Yes" or "No" judgment, updating the `exposure_and_outcome` table accordingly. Pairs that have already undergone MR analysis are excluded from subsequent steps.

Further, low-quality MR analyses can yield unreliable results, potentially missing important exposure-outcome pairs. MRAgent can evaluate the quality of these analyses by invoking the STROBE-MR tool [17]. The LLM assesses MR study papers against the 20 key items in the STROBE-MR checklist, recording a 'Yes' or 'No' for each item to indicate completeness. Before running MRAgent, users can specify which checklist items are critical. If an MR analysis lacks these critical items, it is flagged as low-quality. Such exposure-outcome pairs are reintegrated into the next analysis phase, ensuring significant relationships are not overlooked. These assessments are stored in the `exposure_and_outcome` table.

Step 3: Identifying Synonyms

MRAgent employs the UMLS Terminology Services tool to obtain synonyms for exposures and outcomes, eliminating the need for manual searching and substitution. The exposures and outcomes that have not undergone MR analysis are extracted and stored in the `outcome` table. The tool then retrieves synonyms for all risks and diseases listed in the `outcome` table, with synonyms assigned a common sID for identification.

Step 4: Checking for GWAS Data in OpenGWAS

To enhance MRAgent's efficiency, the agent first accesses the GWAS data corresponding to the entries in the `outcome` table using the OpenGWAS GWAS summary datasets tool. If no GWAS data is found through keyword matching, it is marked as FALSE, and the entry is excluded from further processing.

Step 5: Selecting the GWAS ID

Similar to Step 4 in traditional MR methods, the agent uses the keywords from the `outcome` table to extract GWAS descriptions via the OpenGWAS GWAS summary datasets tool. The LLM, leveraging its domain knowledge, selects the GWAS IDs that match the keywords based on the database's trait descriptions and other relevant columns. The output is formatted as a Python list for subsequent operations and saved in the `outcome` table. Steps 4 and 5 in MRAgent correspond to Step 4 in traditional MR methods.

Step 6: Recombining Exposure-Outcome Pairs

The agent reads the `exposure_and_outcome` table and, for each pair, retrieves synonyms from the `outcome` table. It then performs a Cartesian product of the exposure and outcome synonym lists (including the original exposure and outcome), generating combinations of different terms with the same semantic meaning. These combinations are saved back into the `exposure_and_outcome` table, with the same source exposure and outcome marked by a common oeID. Particularly, to cater to diverse research needs, MRAgent includes an option for bidirectional Mendelian analysis. If bidirectional Mendelian analysis is selected during MRAgent instantiation, this step will duplicate the exposure and outcome pairs, swap their positions, and store them in the `exposure_and_outcome` table. This approach increases the robustness of the analysis by considering all possible synonyms.

Step 7: Checking for Previous MR Analyses Again

Given the newly generated exposure-outcome pairs with different synonyms, the agent rechecks whether any of these pairs have undergone MR analysis. If any synonym combination has been previously analyzed, all combinations with the same semantic meaning are excluded from further analysis. This step is implemented identically to Step 2, and the STROBE-MR tool [17] can similarly be invoked to assess the quality of the MR analyses.

Step 8: Screening Final Exposure-Outcome Pairs for MR

The agent filters the `exposure_and_outcome` table to identify pairs that have not undergone MR analysis and have corresponding GWAS data in the OpenGWAS database. These pairs are saved in the `run` table. This step allows for more refined control over the workflow, enabling researchers to review and verify the pairs before proceeding with the MR analysis.

Step 9: Conducting MR Analysis and Interpreting the Results

Similar to traditional methods, MRAgent will first revisit the OpenGWAS tool to ensure that the GWAS data for exposures and outcomes are derived from the same population. Any matches involving different populations are discarded to maintain the validity of the analysis.

Following this verification, MRAgent employs the TwoSampleMR Tool to conduct MR computations, which include MR or MR-MOE methods. These computations are based on the pairs stored in the `run` table, with their corresponding GWAS IDs stored in the `outcome` table. Each exposure and outcome may correspond to multiple GWAS IDs, and all possible pairwise combinations of these GWAS IDs for exposures and outcomes are matched for MR analysis. Further, MRAgent can optionally invoke MRlap Tool to correct for biases such as sample overlap and weak instruments.

The LLM reads the results from each TwoSampleMR Tool computation and generates individual reports for each exposure-outcome pair. Considering the fixed format of MRlap results, reports for MRlap are generated through logical inference to conserve LLM resources. The report generation template is provided in Appendix III. Additionally, the LLM provides a summary report that consolidates the findings from all individual analyses for a given exposure-outcome pair.

Results

Evaluation of MRAgent

In the development of MRAgent, an LLM has been employed to automate tasks traditionally performed by human researchers, such as literature review, process control, and result analysis. To validate the effectiveness of this approach, we rigorously tested the steps in which the LLM is deeply involved, including Identifying the Exposure and Outcome, Evaluating MR analyses quality, Checking Previous MR Analyses, Selecting the GWAS ID, and Conducting MR Analysis and Interpreting the Results. For the evaluation, we selected both SOTA closed-source and open-source large models and compared their performance against that of human experts. The closed-source models, including GPT-3.5-Turbo (GPT-3.5-Turbo-0125) [22], GPT-4-Turbo (GPT-4-0125-preview) [6], and Claude-3-opus (Claude-3-opus-20240229) [23], were accessed via API connections. The open-source models, including Qwen-max

(Qwen-max-0403) [24], mixtral:8x22b [25], llama3:8b [26, 27], and llama3:70b [26, 27], were run on a GPU server equipped with A100 GPUs. Comparing multiple LLMs helps in selecting the most cost-effective model within an acceptable performance range. We also compared different prompts for interpreting MR results. Finally, we recorded the execution time for different steps and compared it with the time taken by humans.

Currently, evaluation methods for LLMs are diverse and can be broadly categorized into automatic evaluation and human evaluation [28]. In this study, human evaluation was conducted by PhD students and professors from Macau, all of whom possess professional expertise in the medical field. We designed specific evaluation methods tailored to different steps, as detailed below:

Evaluation of identifying the exposure and outcome

MRAgent can automatically read papers from PubMed. We used keywords such as "Alzheimer," "Lung cancer," and "Chronic kidney disease" to fetch the titles and abstracts of the 50 most recent papers from PubMed. These were then fed into different LLM models, which interpreted the articles and extracted the exposures and outcomes. As a standard reference for extracting exposures and outcomes, human experts also conducted the same task.We designed both Human Evaluation and Automatic Evaluation methods to compare the LLM output with the human standard output.

For the Human Evaluation method, human experts labeled each pair of exposures and outcomes identified by different LLMs. The labeling process followed a decision tree, as shown in Fig. 3. Each exposure-outcome pair could be labeled as "Label A: Relevant to the article content and related to the keyword," "Label B: Relevant to the article content but unrelated to the keyword," or "Label C: Not relevant to the article content." As shown in Fig. 4, an example is provided where an LLM interprets a paper [29] and extracts exposures and outcomes, which are then labeled by humans into three categories. We then tallied the number of each label for each model.

For the Automatic Evaluation method, we designed a similarity algorithm based on SimCSE, with pseudocode as shown in Algorithms 1 and 2. This method objectively describes the semantic gap between the LLM-extracted exposures and outcomes and those extracted by humans.

Algorithm 1 LIST-PREPROCESS

Input: *hum_list*, *llm_list*
Output: *hum_list_new*, *llm_list_new*, *mask*

NEW EMPTY LIST *hum_list_new*
NEW EMPTY LIST *llm_list_new*
NEW EMPTY LIST *mask*
for $i = 0$ **to** LENGTH(*hum_list*) **do**
  if *hum_list*[i] $\neq$ [] **and** *llm_list*[i] $\neq$ [] **then**
    APPEND*hum_list_new* **WITH** *hum_list*[i]
    APPEND*llm_list_new* **WITH** *llm_list*[i]
    APPEND*mask* **WITH** 1
  else if (*hum_list*[i] $\neq$ [] **and** *llm_list*[i] $=$ []) **or** (*hum_list*[i] $=$ [] **and** *llm_list*[i] $\neq$ []) **then**
    APPEND*hum_list_new* **WITH** *hum_list*[i]
    APPEND*llm_list_new* **WITH** *llm_list*[i]
    APPEND*mask* **WITH** 0
  end if
end for
return (*hum_list_new*, *llm_list_new*, *mask*)

Algorithm 2 SimCSE-SIMILARITY

Input: *list1*, *list2*
Output: *similarity_mean*

(*hum_list*, *llm_list*, *mask*) ← LIST-PROCESS(*list1*, *list2*)
similarities ← SIMILARITY(*hum_list*, *llm_list*)
for each *index* i **from** 0 **to** LENGTH(*similarities*) **do**
  if *mask*[i] = 0 **then**
    similarities[i] ← 0
  end if
end for
similarity_mean ← MEAN(*similarities*)
return *similarity_mean*

The results of these evaluations are summarized in Table 1. Human evaluation results indicate that both GPT-4-Turbo and Qwen-max models identified the highest number of correctly labeled exposure-outcome pairs, with 43 pairs marked as "Label A" each, and an equal number of pairs in "Label B" and "Label C." Human experts identified a total of 98 pairs, serving as the benchmark for comparison. Notably, the open-source model llama3:70b showed performance close to the closed-source models GPT-4-Turbo and Qwen-max. Given that open-source models do not require real-time usage fees like closed-source models, llama3:70b presents a cost-effective option. For the Automatic Evaluation, the GPT-4-Turbo model achieved the highest SimCSE similarity score of 0.5285, indicating the closest semantic alignment with human-extracted pairs. Overall, the GPT-4-Turbo model demonstrated superior reliability in identifying exposure-outcome pairs.

Evaluation of checking previous MR analyses

In Steps 2 and 7, MRAgent determines whether the Exposure-Outcome Pairs have already undergone MR analysis. The LLM outputs are formatted as "Yes" or "No," effectively making this a binary classification task. We manually selected 40 Exposure-Outcome Pairs, confirming through meticulous document searches that 20 pairs had undergone MR analysis while the remaining 20 had not. Using the manually labeled results as ground truth, we conducted an Automatic Evaluation to calculate the accuracy as follows:

$$Accuracy = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \quad (1)$$

As illustrated in Table 2, the experimental results demonstrate that the Qwen-max model achieved the highest accuracy, scoring 0.825. Notably, the open-source models llama3:70b and mixtral:8x22b performed on par with the GPT-4-Turbo model, each attaining an accuracy of 0.8. All models exhibited strong performance, with accuracies exceeding 0.7, indicating that the task of determining whether MR analysis has been conducted is relatively straightforward.

Evaluation of evaluate MR analyses quality

In Steps 2 and 7, MRAgent is capable of reading articles related to completed MR analyses and generating a STROBE-MR checklist [17], where each item receives a binary outcome.To assess the performance of this feature, we selected four published papers and manually evaluated them to produce a STROBE-MR checklist. We then compared these manual evaluations with the results

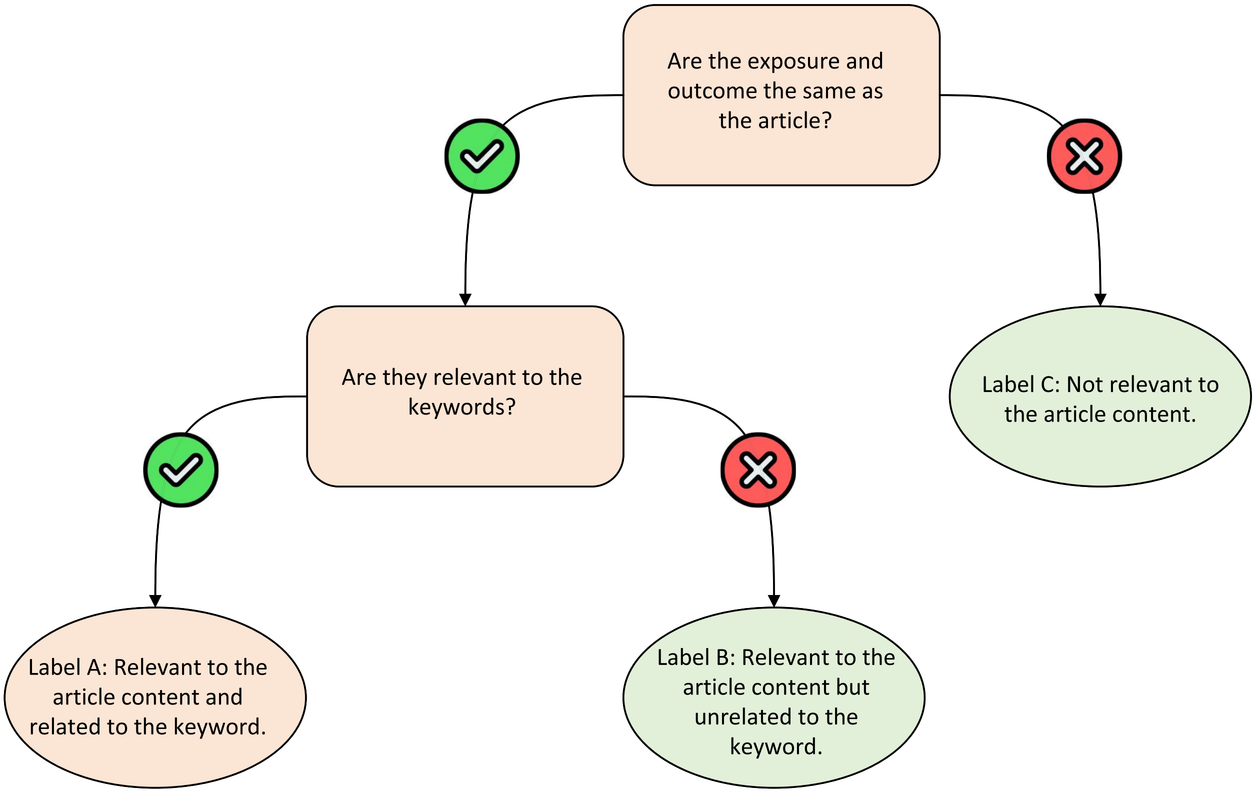

Figure 3. The decision tree of labeling.

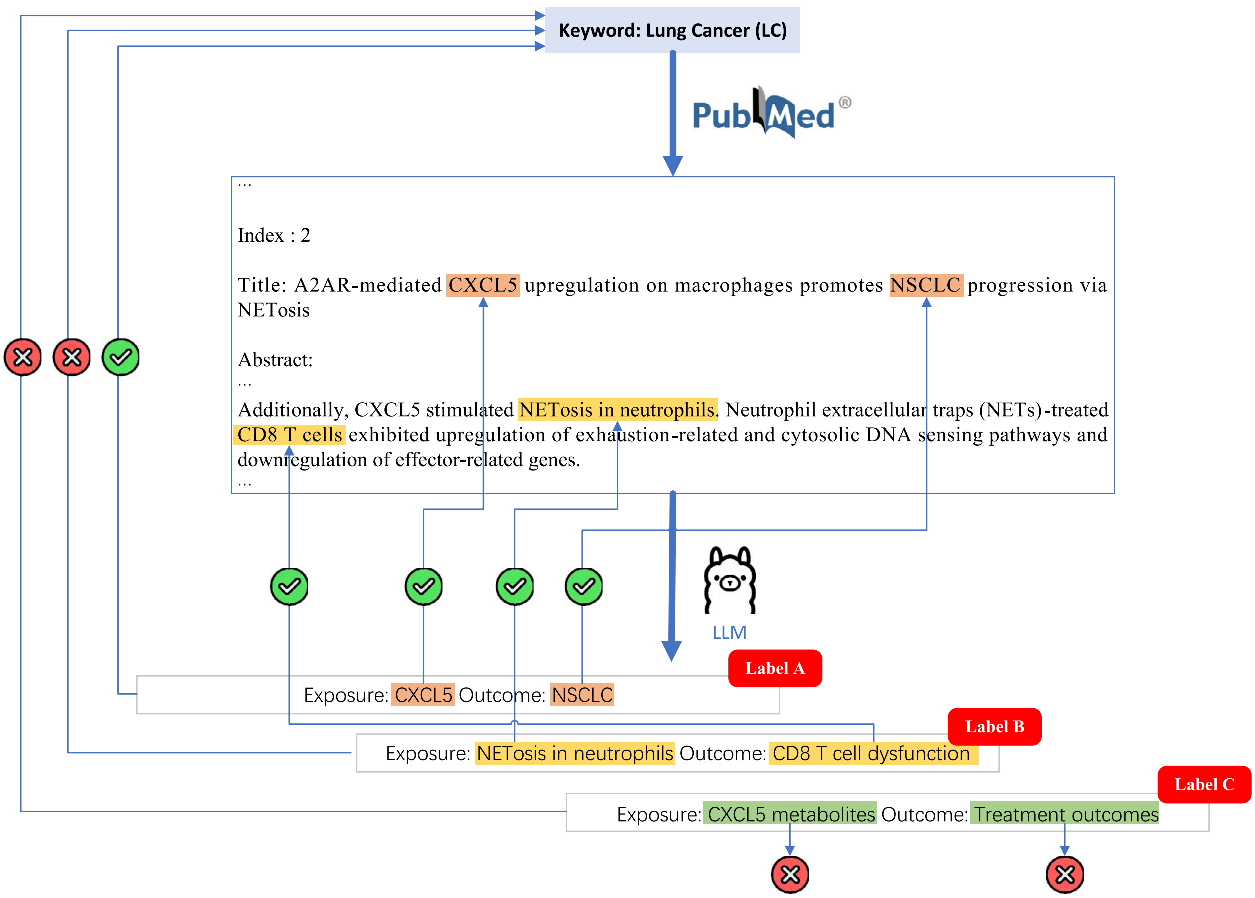

Figure 4. Example of labeling.

Table 1. Num. of exposure-outcome pairs, each label and SimCSE similarity

Model	Num. of exposure-outcome pairs	Label A	Label B	Label C	SimCSE similarity
GPT-3.5-Turbo	19	12	2	5	0.1765
GPT-4-Turbo	65	43	12	10	0.5285
Claude-3-opus	21	14	6	1	0.2190
Qwen-max	66	43	14	8	0.4521
mixtral:8x22b	41	19	11	11	0.3261
llama3:8b	40	23	9	8	0.3448
llama3:70b	64	40	14	10	0.3970
human	98	–	–	–	–

Table 2. Accuracy of checking previous MR analyses

Model	Accuracy
GPT-3.5-Turbo	0.775
GPT-4-Turbo	0.800
Claude-3-opus	0.775
Qwen-max	0.825
mixtral:8x22b	0.800
llama3:8b	0.700
llama3:70b	0.800

Table 3. Accuracy of STROBE-MR checklist

Model	Accuracy
GPT-3.5-Turbo	Out of context length
GPT-4-Turbo	0.750
Claude-3-opus	Formatting failure
Qwen-max	Out of context length
mixtral:8x22b	0.540
llama3:8b	Out of context length
llama3:70b	Out of context length

generated by the LLM. This comparison constitutes a binary classification task, and we evaluated the performance using accuracy, calculated as described in Equation (1).

For each LLM model, we calculated the average accuracy across all articles, with the results presented in Table 3. Among the models tested, only GPT-4-Turbo and mixtral:8x22b successfully completed the evaluation. GPT-4-Turbo achieved the highest accuracy, reaching 0.750. GPT-3.5-Turbo, Qwen-max, llama3:8b, and llama3:70b encountered issues related to context length, which prevented them from effectively completing the task. Additionally, Claude-3-opus experienced a formatting failure, as it was unable to output the required JSON format.

Evaluation of selecting relevant GWAS

In Step 5, MRAgent selects appropriate GWAS IDs from the OpenGWAS database. We chose 30 diseases covering various body systems, and manually selected suitable GWAS IDs from the OpenGWAS database for each disease. The same task was also performed by Agents based on different LLMs. For each disease, we conducted an Automatic Evaluation to calculate precision, recall, and F1 score. We consider two sets: A and B. Set A represents the collection of GWAS IDs selected by experts, while set B denotes the GWAS IDs output by the LLM. The intersection of these two sets contains the GWAS IDs that are common to both A and B,

$$C = A \cap B \tag{2}$$

The precision and recall are calculated as follows:

$$Precision = \frac{|C|}{|B|} \tag{3}$$

$$Recall = \frac{|C|}{|A|} \tag{4}$$

The F1 score, which is the harmonic mean of precision and recall, provides a single metric that balances both aspects. The F1 score is calculated as follows:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{5}$$

We then averaged the precision, recall, and F1 scores across all 30 diseases. As illustrated in Table 4, the analysis of the results indicates that Qwen-max achieved the highest balance between precision and recall, with a precision of 0.8152, a recall of 0.7891, and an impressive F1-score of 0.7675. This suggests that Qwen-max is the most effective model in maintaining both high accuracy and completeness in identifying relevant GWAS IDs. GPT-4-Turbo demonstrated strong performance with an F1-score of 0.7203. This model effectively balances precision and recall, making it a reliable choice for identifying GWAS IDs. Claude-3-opus exhibited the highest precision at 0.9767, indicating a high accuracy in the GWAS IDs it identified. However, its recall was relatively lower at 0.5797, resulting in an F1-score of 0.6755. This implies that while Claude-3-opus is very precise, it may miss a significant number of relevant GWAS IDs. On the other hand, mixtral:8x22b, llama3:8b, and llama3:70b performed poorly in comparison, with notably lower precision, recall, and F1-scores. Their lower performance suggests that these models are less effective in accurately and comprehensively identifying relevant GWAS IDs.

Evaluation of interpreting the MR results

The TwoSampleMR Tool offers both standard MR and MR-MOE modes, and MRAgent can perform detailed analyses of the results and generate reports. We selected 8 exposure-outcome pairs for MR analysis, with 4 pairs for each mode, including both significant and non-significant causal relationships. Different LLMs generated reports based on the results, and human experts also wrote reports for the same results as a reference. We designed both Human Evaluation and Automatic Evaluation methods to compare the LLM-generated reports with human-standard reports.

Table 4. Evaluation of selecting relevant GWAS

Model	Precision Avg	Recall Avg	F1-Score Avg
GPT-3.5-Turbo	0.8940	0.5509	0.6123
GPT-4-Turbo	0.8864	0.6950	0.7203
Claude-3-opus	0.9767	0.5797	0.6755
Qwen-max	0.8152	0.7891	0.7675
mixtral:8x22b	0.5397	0.3526	0.4020
llama3:8b	0.2972	0.1284	0.1614
llama3:70b	0.5909	0.4828	0.5133

The Human Evaluation method was designed as a single-blind experiment, where reports generated by different LLMs and humans were randomly shuffled and anonymized. Another set of researchers was asked to score all the anonymized reports based on the MR analysis results using a Likert-style rating scale, as shown in Table 5. The Automatic Evaluation method was also based on SimCSE [15], calculating the cosine similarity between the LLM-generated reports and the reports written by human experts.

For the standard MR mode, as shown in Table 6, Qwen-max achieved the highest average scores across all four questions and the highest SimCSE similarity score, indicating superior performance in automatic evaluation metrics and semantic similarity to human expert reports. This was followed closely by Claude-3-opus and GPT-4-Turbo, both of which also demonstrated high levels of accuracy and similarity to human-generated reports. In contrast, the open-source model mixtral:8x22b performed the best among the open-source options but still showed a significant gap compared to the closed-source models. In the single-blind scoring, the reports written by human experts did not surpass those generated by LLMs, indicating that LLMs have reached a human-level performance in MR report writing tasks.

For the MR-MOE mode, as shown in Table 7, Claude-3-opus achieved the best human evaluation scores, while GPT-4-Turbo exhibited the highest SimCSE similarity score. However, the open-source models performed poorly in this mode. Overall, the scores for interpreting MR-MOE were lower across all models compared to the standard MR mode. This could be due to the increased complexity of MR-MOE results and its less frequent usage, leading to fewer training examples for the models. Notably, the reports written by human experts did not exceed the scores of those generated by LLMs in this mode either.

Overall, the results indicate that LLM-generated reports have achieved a level of quality comparable to that of human experts. The outstanding performance of advanced LLMs in both MR and MR-MOE modes suggests that these models can serve as valuable tools for facilitating and automating complex genetic epidemiological studies.

Comparison of prompt strategies for interpreting MR results

MRAgent employs an LLM to interpret results from both MR and MR-MOE methods within the TwoSampleMR Tool. To enhance the accuracy and coherence of the generated reports, we explored various prompt strategies. These included: Zero Shot, where only the results are provided for report generation; Few Knowledge, which involves supplying a small amount of additional information to guide the analysis; One Shot, offering a single example report as a template; and One Shot and Few Knowledge, combining an example report with supplementary guiding information. Additionally, we evaluated the Zero Shot CoT (Chain of Thought) strategy [30], which incorporates the prompt "Let's think step by step" to encourage a structured reasoning process, as well as Zero Shot CoT and Few Knowledge, which enhances this approach with additional guiding information. Our evaluation aimed to identify the most effective strategy for improving the LLM's interpretative capabilities in processing MR results.

To assess the effectiveness of these prompt strategies, we applied both MR and MR-MOE computations to six distinct exposure-outcome pairs. For each computation, reports were generated using the aforementioned prompt strategie, details of prompt are provided in Appendix 4s. The evaluation employed the GPT-4o model [31] to generate the reports. Subsequently, we invited human experts to evaluate and rank the quality of these reports. To ensure anonymity and prevent bias during evaluation, each strategy was represented by a number rather than its descriptive name. The experts' evaluations and rankings were systematically recorded, as shown in Table 8.

The analysis of the prompt strategies revealed that the "One Shot and Few Knowledge" approach was the most effective, achieving the highest rank. This was followed by "Few Knowledge" and "One Shot," which ranked second and third, respectively, while "Zero Shot" performed the worst. These findings suggest that the GPT-4o model lacks inherent knowledge related to MR and requires additional information to generate accurate reports. When provided with sufficient information, the model demonstrated a strong ability to produce coherent and insightful analyses, highlighting its robust analytical capabilities. Furthermore, the "Zero Shot CoT" strategy yielded relatively poor results, further indicating that GPT-4o lack of MR-specific knowledge is insufficient to support a step-by-step reasoning process through the Chain of Thought approach. This underscores the necessity of supplementing the model with targeted information to enhance its interpretative performance in MR analyses.

In constructing the final MRAgent, we adopted the "One Shot and Few Knowledge" strategy for its prompt, leveraging its proven effectiveness in generating comprehensive and accurate reports.

Table 5. Likert-style rating scale

Q1. The data described in the report is the same as the actual MR results?				
1-Completely different	2-Somewhat different	3-Average	4-Somewhat similar	5-Completely similar
Q2. Is the analysis in the report biologically meaningful?				
1-Completely inconsistent	2-Somewhat inconsistent	3-Average	4-Somewhat consistent	5-Completely consistent
Q3. Does the report contain sufficiently detailed analysis (including sufficiently detailed explanations of biological significance and presentation of detailed data)?				
1-Not detailed enough	2-Somewhat insufficiently detailed	3-Average	4-Fairly detailed	5-Very detailed
Q4. Is the final result given in the report accurate?				
1-Completely inaccurate	2-Mostly accurate	3-Average	4-Mostly accurate	5-Completely accurate

Table 6. Evaluation of MR mode report

Model	Q1 Avg	Q2 Avg	Q3 Avg	Q4 Avg	SimCSE similarity
GPT-3.5-Turbo	5.00	4.75	3.00	4.75	0.7647
GPT-4-Turbo	5.00	5.00	4.75	5.00	0.7876
Claude-3-opus	5.00	4.75	4.75	5.00	0.8031
Qwen-max	5.00	5.00	5.00	5.00	0.8501
mixtral:8x22b	4.00	4.00	2.75	3.50	0.7998
llama3:8b	1.25	1.50	3.00	1.00	0.7996
llama3:70b	2.75	2.75	2.00	2.75	0.7666
human	4.50	4.75	4.50	4.75	–

Table 7. Evaluation of MR-MOE mode report

Model	Q1 Avg	Q2 Avg	Q3 Avg	Q4 Avg	SimCSE similarity
GPT-3.5-Turbo	4.50	3.75	3.75	4.50	0.7752
GPT-4-Turbo	4.25	4.25	3.75	4.00	0.8577
Claude-3-opus	4.75	4.75	4.00	4.25	0.8369
Qwen-max	4.00	4.00	4.00	4.00	0.7945
mixtral:8x22b	3.25	3.25	2.25	2.50	0.7137
llama3:8b	1.75	2.25	1.50	1.50	0.6479
llama3:70b	1.75	2.00	1.50	2.00	0.6658
human	4.25	4.50	4.00	4.00	–

Table 8. Evaluation of prompt strategies for MR and MR-MOE

Index	Prompt strategy	MR	MR-MOE	Rank
1	Zero shot	Describes the results of each method but lacks rigor in the analysis conclusions	Provides basic analysis but lacks in-depth discussion on heterogeneity and pleiotropy	6
2	Few knowledge	Main methods and results are clearly interpreted, but sensitivity analysis interpretation is insufficient	Offers comprehensive analysis and interpretation of results, depth in heterogeneity analysis and pleiotropy testing is close to Index 4	2
3	One shot	Describes the results of each method but lacks rigor in the analysis conclusions	Covers major MR methods but lacks depth in sensitivity analysis	3
4	One shot and few knowledge	Provides detailed sensitivity analysis and interpretation of heterogeneity and pleiotropy, with a deep understanding of methods, most professional description among these	Offers thorough explanation of MR methods and sensitivity analysis, demonstrates heterogeneity and pleiotropy testing, conclusions are rigorous and reliable	1
5	Zero shot CoT	Describes the results of each method but lacks rigor in the analysis conclusions	Provides basic analysis but lacks in-depth discussion on heterogeneity and pleiotropy	5
6	Zero shot CoT and few knowledge	Describes the results of each method but lacks rigor in the analysis conclusions	Provides basic analysis but lacks in-depth discussion on heterogeneity and pleiotropy	4

Execution time analysis of MRAgent

To gain insights into the efficiency of MRAgent, we recorded the execution time for each step in the workflow. The Python `time` module was employed to measure the duration of each function. The evaluation employed the GPT-4o model [31]. Given that each function is typically executed multiple times during a single run of MRAgent, Table 9 presents the average execution time for each step. For steps involving the use of an LLM, we also provide an approximate duration for humans performing the same tasks [32]. This comparison highlights the potential time savings and efficiency gains achieved by utilizing MRAgent.

The execution time analysis highlights MRAgent's remarkable efficiency compared to human experts across several critical tasks. For instance, extracting exposure and outcome information from a paper takes human experts over 5 min to review abstracts and key sections, whereas MRAgent completes this task in under one second. Similarly, determining if an exposure-outcome pair has undergone MR analysis requires extensive literature review, taking experts more than 30 min, while MRAgent accomplishes this in just 2.883 s by leveraging the LLM. When evaluating the quality of published MR analyses using the STROBE_MR checklist, human experts need over 20 min for a thorough

Table 9. Execution time for each function

Step	Function	Average time (s)	Human time (min)
1	Extracting exposure and outcome from a paper	0.964	>2
2,7	Checking if an exposure-outcome pair has undergone MR analysis	2.883	>30
2,7	Analyzing the quality of published MR analyses using the STROBE_MR checklist	8.937	>20
3	Finding synonyms for exposure-outcome pairs	1.905	–
4	Checking for GWAS data in OpenGWAS	1.965	–
5	Selecting the GWAS ID	0.676	1-5
6	Recombining exposure-outcome pairs	0.370	–
8	Screening final exposure-outcome pairs for MR	0.024	–
9	Running MR using the TwoSampleMR R package	17.762	–
9	Analyzing MR Results	7.728	>30
9	Further correcting results using MRlap	1296.974	–

reading, whereas the LLM completes this evaluation in 8.937 s. Additionally, selecting the GWAS ID, which varies between 1 and 5 min for experts depending on information availability, is performed by MRAgent in less than a second. After obtaining MR computational results, human experts require over 30 min to analyze results and draft reports, whereas MRAgent, utilizing the LLM, completes this in just 7.728 s. Collectively, these comparisons underscore MRAgent's ability to significantly reduce the time required for MR analysis, demonstrating substantial efficiency gains.

It is note that further correcting results using MRlap requires downloading comprehensive GWAS data locally and involves substantial computation, resulting in slower processing times.This step is optional, and considering operational efficiency, it is recommended to only choose this step when necessary.

Full process proof-of-concept cases

To validate the feasibility of MRAgent in conducting causal knowledge discovery, we conducted a comprehensive proof-of-concept study. The overall process, depicted in Fig. 5, envisions a scenario where a physician seeks to explore the causes associated with back pain. The physician inputs "Outcome: back pain" into the system, selects the Knowledge Discovery mode, sets the MR method to standard MR, and opts for the GPT-4o LLM model. Additionally, the physician must provide an OpenAI API key and an OpenGWAS token.

Initially, MRAgent reviews 300 relevant research articles to identify exposures and outcomes related to back pain. Upon execution, the system populates the `exposure_and_outcome` table with all identified pairs from the literature, facilitating clinical diagnosis and treatment strategies for low back pain. MRAgent then automatically performs MR analysis on all exposure-outcome pairs in the table that have not been previously analyzed, considering all possible combinations of their respective GWAS IDs.

For example, as illustrated in Fig. 5, MRAgent conducts multiple analyses using various GWAS IDs for the exposure "spondylolysis" (GWAS ID: finn-b-M13_SPONDYLOLISTHESIS) and the outcome "low back pain" (GWAS ID: finn-b-M13_LOWBACKPAIN). The LLM model delivers a detailed analysis of the standard MR results, providing strong evidence for a causal relationship between genetically predicted spondylolysis and an increased risk of chronic low back pain.

Furthermore, as depicted, MRAgent compiles all reports, summarizing the MR analyses conducted across different GWAS IDs. Consistent significant causal effects were observed for the same exposure-outcome pair of spondylolysis and low back pain across various GWAS IDs.

MRAgent executes multiple MR analyses related to back pain for diverse exposures and outcomes, generating several reports. Ultimately, it concludes a causal relationship between spondylolysis and low back pain, as well as between osteoarthritis and back pain.

Discussion

MRAgent contributes to the field of causal knowledge discovery in medical research. By utilizing LLMs, MRAgent automates the process of identifying exposure-outcome pairs from scientific literature and conducting MR analyses. This tool operates in two modes: "Knowledge Discovery," which autonomously extracts potential causal relationships from literature, and "Causal Validation," which allows researchers to input predefined exposure-outcome pairs for analysis. MRAgent's ability to perform large-scale causal inference using Genome-Wide Association Study (GWAS) data provides researchers and clinicians with a powerful tool for exploring and validating causal relationships in complex diseases.

Our experiments demonstrate that LLMs can effectively replace human experts in tasks such as literature review, data analysis, and causal reasoning in the domain of bioinformatics. MRAgent can effectively accomplish the task of causal knowledge discovery related to diseases. Among the models tested, GPT-4-Turbo emerged as the most effective, with Qwen-Max also performing well, albeit limited by its shorter context length, which hinders its ability to fully evaluate MR quality. Open-source models, while cost-effective, exhibited weaker performance. From a temporal perspective, MRAgent significantly accelerates the knowledge discovery process, reducing the time required for MR analysis compared to manual methods. The successful implementation of MRAgent in proof-of-concept cases, such as exploring the causal factors of back pain, underscores its potential to significantly enhance the efficiency and accuracy of causal inference in medical research.

The biomedical implications of MRAgent are profound. In clinical and public health contexts, the ability to swiftly identify causal relationships can inform prevention strategies and therapeutic interventions. Traditionally, studies often provide statistical validation for specific exposure-outcome pairs or document isolated case studies, offering limited causal evidence. However, MRAgent can provide stronger causal insights in related research, enhancing our understanding of disease mechanisms. Moreover,

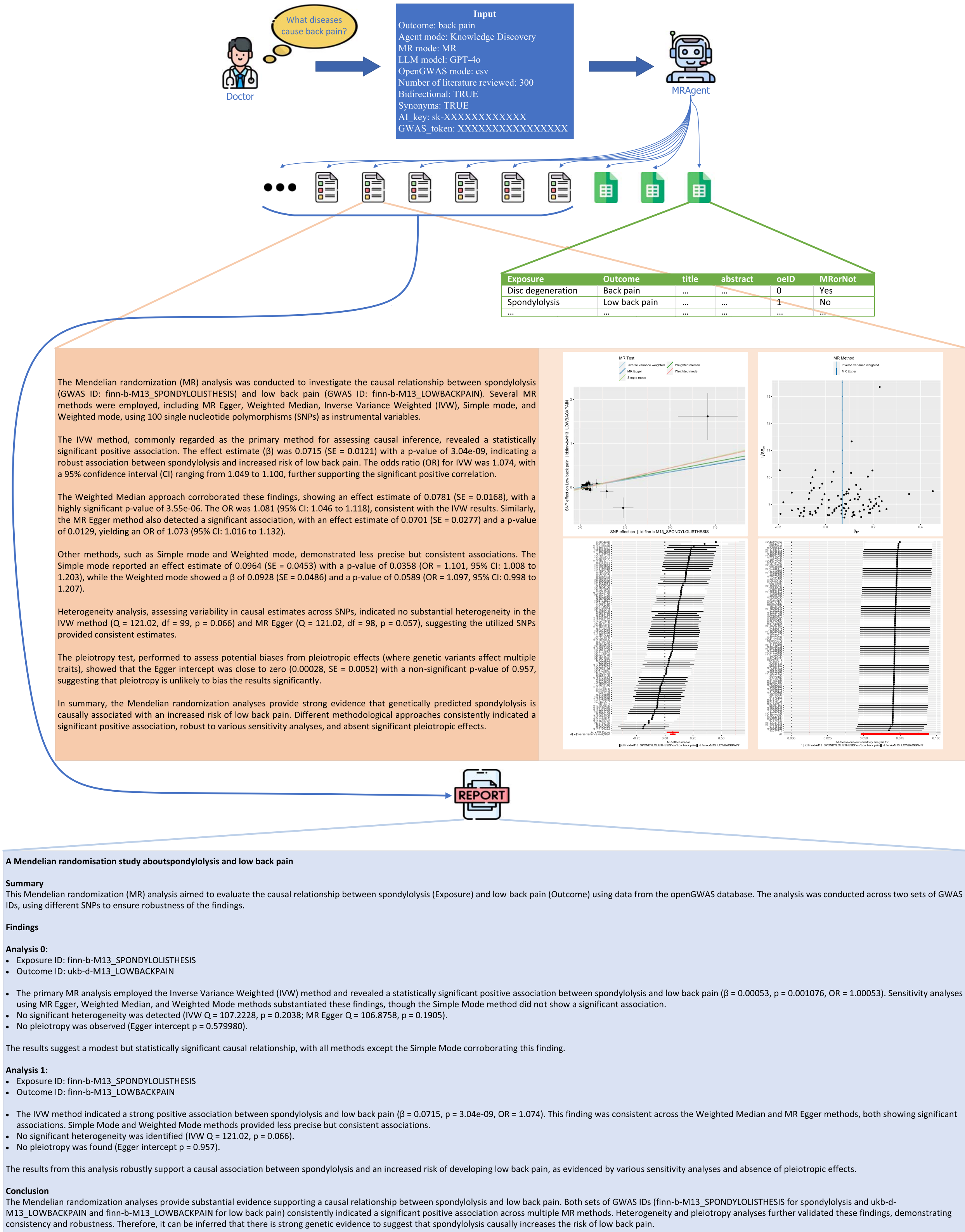

Figure 5. Proof-of-Concept case.

the rapid pace of scientific advancement and the vast volume of medical literature pose challenges for clinicians, who may struggle to stay informed within their field. MRAgent assists by quickly extracting disease-related causal knowledge from the latest literature. Additionally, MRAgent provides a systematic approach to uncovering causal relationships in complex and rare diseases, enabling ordinary clinicians to quickly and comprehensively understand intricate disease mechanisms. This capability ultimately improves patient care and advances medical research.

Conclusion

In this study, we presented MRAgent, an innovative automated agent designed to leverage LLMs for advancing causal knowledge discovery in disease research through MR. MRAgent excels in autonomously extracting exposure-outcome pairs from scientific literature and conducting comprehensive MR analyses, thereby providing robust causal insights. Our experimental results underscore the efficacy of MRAgent in automating tasks that traditionally require human expertise, such as literature review and data interpretation. This automation significantly reduces the time needed for causal inference. MRAgent represents an advancement in the field of disease research, offering clinicians and researchers a powerful tool for efficiently exploring and validating causal relationships in complex diseases.

Future work could focus on integrating additional causal inference algorithms, which would further enhance MRAgent's capabilities, allowing for even more efficient and comprehensive causal knowledge discovery.

Key points

- Establishing causality in medical research is vital for developing effective interventions and diagnostic tools. However, traditional methods often rely on the pre-identification of exposure-outcome pairs, which can be difficult to ascertain, creating challenges for clinicians seeking to investigate causal factors of specific diseases.
- We present MRAgent, an automated agent using LLMs to enhance causal discovery in disease research via MR, autonomously identifying exposure-outcome pairs from literature.
- Evaluations of various LLMs demonstrate MRAgent's effectiveness, highlighted by a proof-of-concept case showcasing its comprehensive workflow and large-scale causal analysis capabilities.
- MRAgent advances medical research by providing a powerful tool for researchers and clinicians to systematically explore and validate causal relationships, facilitating informed decision-making in complex diseases.

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Author contributions

W.X., G.L., and K.Z. conceived and designed the study. W.X., G.L., W.M., X.Z., Y.L., A.X., J.L., Z.L., and K.Z. conducted the experiments. W.X. and J.W. wrote the initial draft of the manuscript. K.L. managed the project and revised the final manuscript.

Supplementary data

Supplementary data is available at *Briefings in Bioinformatics* online.

Conflict of interest: None declared.

Data availability

The codes developed in this study are openly accessible on GitHub at https://github.com/xuwei1997/MRAgent. The code has been packaged and released on PyPI, allowing for easy installation using pip by following the instructions provided at https://pypi.org/project/mragent/. The web demo is hosted on Hugging Face and can be accessed at https://huggingface.co/spaces/xuwei1997/MRAgent. During the evaluation phase, the test data generated by the code has been uploaded to Zenodo and can be accessed via https://doi.org/10.5281/zenodo.14184396.

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Problem Solving Protocol